
REPORT FOR PHENOBARBITAL VERSUS BENZODIAZEPINE-ONLY TREATMENT FOR ALCOHOL WITHDRAWAL IN THE EMERGENCY DEPARTMENT

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ABSTRACT

This retrospective study of 1,267 emergency department encounters for alcohol withdrawal compares outcomes between phenobarbital (PHB) and benzodiazepine-only (BZD) treatment strategies. Descriptive analyses identified significant differences between groups in baseline characteristics, including CIWA score, ethanol level, gender, and race, as well as in treatment and outcomes. Regression models, supplemented with LASSO and Elastic Net, were used to evaluate associations between treatment, dosing, and clinical outcomes. Across models, treatment group, cumulative BZD dose, and route of administration were the strongest predictors of patient disposition. PHB use was associated with greater improvement in CIWA scores but a substantially higher likelihood of hospital admission, likely reflecting confounding by indication. Higher cumulative BZD dosing was consistently associated with increased odds of admission and poorer symptom trajectories. These findings represent statistical associations rather than causal effects, and a randomized, double-blind controlled trial would be necessary to establish causal relationships.

1 Introduction

This study analyzes a retrospective cohort of Emergency Department (ED) encounters for alcohol withdrawal syndrome between June 2020 and July 2025. The analytic dataset includes 1267 encounters and captures patient-level demographics, treatment exposures, and outcome measures with missing values. The primary objective is to compare disposition between treatment regimens, defined as phenobarbital (PHB) therapy (with or without adjunctive benzodiazepines (BZD)) versus BZD-only therapy. To account for treatment intensity, regimens are further stratified into high- and low-dose subgroups using predefined thresholds as listed in Table 1 below.

Table 1: Subgroup Definition

Category	Definition
BZD High Dose	Patient did not receive phenobarbital, and the first benzodiazepine dose was > 20 mg diazepam equivalents.
BZD Low Dose	Patient did not receive phenobarbital, and the first benzodiazepine dose was ≤ 20 mg diazepam equivalents.
PHB High Dose	Patient's first phenobarbital dose was ≥ 260 mg.
PHB Low Dose	Patient's first phenobarbital dose was < 260 mg.

PHB = phenobarbital group; BZD = non-phenobarbital or benzodiazepine only group.

Descriptive analyses are conducted using distribution-appropriate summaries, and group differences are evaluated using Chi-square, Fisher's exact, or Two-sample t-test as appropriate. We further take advantage of regression models to better understand the associations between variables. The primary outcomes, ED disposition which is a categorical variable with four levels, are studied using multivariable logistic regression, adjusting for potential confounders including age, race, gender, vital signs, ethanol levels, and cumulative medication dosing. Binary categorical outcomes like readmitted rate within 3 days is interpreted with logistic regression. Adjusted odds ratios are used in all logistic regressions to quantify effect sizes. Secondary outcomes, including continuous measures such as change of CIWA (Clinical Institute Withdrawal Assessment for Alcohol) score in ED and hospital length of stay, are analyzed using linear regression models.

Given the high dimensionality and potential collinearity among clinical and treatment-related variables, regularization, including Lasso and Elastic net, is employed as a complementary modeling strategy for variable selection and coefficient stabilization. Elastic net balances L1 (Lasso) and L2 (Ridge) penalties to identify a parsimonious set of predictors while mitigating overfitting. Model tuning is conducted via cross-validation. Different subsets of variables are chosen from Lasso and Elastic net, and they are compared by the Akaike information criterion (AIC) of their corresponding simplify models fitting against patient's disposition. Although AIC is not suitable for comparing models across different datasets, a lower AIC indicates a better model fit when comparing models fitted to the same dataset. The penalty term from AIC discourages overfitting, which is desired because increasing the number of parameters in the model almost always improves the goodness of the fit.

This integrated modeling framework—combining traditional regression with penalized methods—provides a rigorous approach to estimating the association between treatment regimen, dosing intensity, and clinical outcomes while addressing confounding, multicollinearity, and model stability in observational ED data.

2 Dataset Preprocessing Description and Descriptive Statistics

Our dataset includes 1,267 observations, with each row representing a single ED encounter. Patients may have contributed multiple encounters over time; however, the dataset does not include patient identifiers or indicators to distinguish repeated encounters from the same individual.

Ethanol levels (EtOH) were harmonized from mixed character and numeric formats: string values such as “<10” were recoded to a numeric value of 10, while all other values remains numeric. In parallel, a binary indicator variable (EtOH <10) was created to capture whether ethanol levels were below the detectable threshold.

Table 2: Racial Distribution in the Dataset.

Race Category	Count (n)	Percentage (%)
White	885	69.85%
Black / African American	221	17.44%
Other Race	124	9.79%
Prefer Not to Answer	14	1.10%
Unknown	9	0.71%
Asian	5	0.39%
American Indian / Alaskan Native	3	0.24%
Asian Indian	2	0.16%
Japanese	2	0.16%
Other Asian	2	0.16%
Total	1,267	100.0%

Percentages are rounded to two decimal places. Categories are sorted by descending frequency.

The racial distribution of the dataset is shown in Table 2, with several categories having relatively small sample sizes. Race categories were consolidated to reduce sparsity and improve model stability: several low-frequency categories (e.g., Asian subgroups and American Indian/Alaskan Native) were grouped into a single “OTHER RACE” category, while ambiguous responses (e.g., “UNKNOWN”, “PREFER NOT TO ANSWER”) were treated as missing.

Variables, summarized using distribution-appropriate statistics and stratified by drug group (BZD vs. PHB), are presented in Table 4 (Baseline), Table 5 (Treatment), and Table 6 (Outcome). Observations with missing values were excluded from analyses and modeling; however, no variables were removed due to missingness, as the highest proportion of missing data among baseline variables was 22%.

Table 3: P-value Evaluation Test

Test	Criteria
Pearson's Chi-Square	Categorical variable with all expected cell counts were at least 5; Yates' correction is employed for variables with two levels.
Fisher's Exact	Categorical variable with any expected cell counts were less than 5.
Independent Two-sample t	Numeric and continuous variables.

The p-values for variables in Table 4, Table 5, and Table 6 comparing distributional differences across drug groups were calculated using the Chi-Square test, Fisher's Exact test, or two-sample t-test, based on the criteria outlined in Table 3. P-values less than 0.05 are conventionally considered significant in statistics and a lower value indicates a stronger significant.

In Table 4, we conducted independent distributional tests to test the group difference between the BZD and PHB for each baseline variables. *First CIWA score in the ED*, ethanol level (*EtOH* and *EtOH* < 10), and *Gender* show statistically significant differences in their distributions between the BZD and PHB groups ($p < 0.001$). Additionally, the distributions of *Race* and *Diastolic BP* differ significantly between the two groups ($p < 0.01$). We also fit a logistic regression to predict the drug group using the baseline variables. Different sets of variables are statistically significant in the logistic regression model and we will be discussing the difference in the next section.

Table 4: Baseline Variables: Gender, Ethanol Level, and First CIWA score in ED show a strong significant difference in the BZD and PHB group.

Variable	BZD (N = 1021)	PHB (N = 246)	p value	Test
Demographics				
Age (years), median (IQR)	46 (35–57)	42 (36–57)	0.914	3
Unknown, no.	1	0		
Race, no. (%)			0.004**	1
Black/ African American	195 (20%)	26 (11%)		
White	698 (70%)	187 (76%)		
Other Race	106 (11%)	32 (13%)		
Unknown, no.	22	1		
Gender, no. (%)			< 0.001***	1
Female	333 (33%)	47 (19%)		
Male	688 (67%)	199 (81%)		
Weight (kg), median (IQR)	79 (68–93)	77 (69–92)	0.361	3
Unknown, no.	81	1		
Testing Results				
First CIWA score in ED, median (IQR)	8 (4–13)	12 (6–17)	< 0.001***	3
Unknown, no.	274	5		
EtOH (mg/dL), median (IQR)	130 (10–260)	35 (10–155)	< 0.001***	3
Unknown, no.	133	4		
EtOH < 10, no. (%)	259 (29%)	100 (41%)	< 0.001***	1
Unknown, no.	132	4		
Systolic BP (mmHg), median (IQR)	138 (124–152)	140 (128–152)	0.406	3
Unknown, no.	63	9		
Diastolic BP (mmHg), median (IQR)	92 (80–103)	95 (85–105)	0.003**	3
Unknown, no.	63	9		

PHB = phenobarbital group; BZD = non-phenobarbital or benzodiazepine only group; CIWA = Clinical Institute Withdrawal Assessment for Alcohol. Significance codes: '***' $p < 0.001$, '**' $p < 0.01$, '*' $p < 0.05$. Test Label: 1 = Pearson's Chi-Square; 2 = Fisher's Exact ; 3 = Independent Two-sample t. The criteria of the evaluation test is listed in Table 3.

In Table 5, we conducted independent distributional tests to test the group difference between the BZD and PHB for each treatment variables. *Subgroup* and *Ordered Route* show statistically significant differences in their distributions between the BZD and PHB groups ($p < 0.001$). Other treatment variables, including *Cumulative PHB*, *Cumulative BZD*, and the indicator of receiving BZD, also differ significantly between the groups. However, these differences mainly reflect the inherent characteristics of the PHB and BZD groups and provide limited additional insight.

Table 5: Treatment Variable: Doesage Subgroup and Ordered Route show a strong significant difference in the BZD and PHB group.

Variable	BZD (N = 1021)	PHB (N = 246)	p value	Test
Cumulative PHB (mg) in ED, median (IQR)	0 (0–0)	194 (130–325)	< 0.001***	3
Cumulative BZD (mg) in ED, median (IQR)	10 (10–23)	20 (0–40)	< 0.001***	3
Recieved BZD, no. (%)	1,021 (100%)	186 (76%)	< 0.001***	1
Subgroup (High Dose), no. (%)	0 (0%)	15 (6.1%)	< 0.001***	2
Ordered Route, no. (%)			< 0.001***	2
Oral	541 (53%)	90 (37%)		
Intramuscular	98 (9.6%)	2 (0.8%)		
Intravenous	382 (37%)	154 (63%)		
Received Sedative Med in ED, no. (%)	546 (53%)	129 (52%)	0.825	

PHB = phenobarbital group; BZD = non-phenobarbital or benzodiazepine only group; BZD = benzodiazepine. Significance codes: '***' $p < 0.001$, '**' $p < 0.01$, '*' $p < 0.05$. Test Label: 1 = Pearson's Chi-Square; 2 = Fisher's Exact ; 3 = Independent Two-sample t. The criteria of the evaluation test is listed in Table 3.

In Table 6, we conducted independent distributional tests to test the group difference between the BZD and PHB for each outcome variables. *Last CIWA score in ED* and *Change of CIWA score in ED* both show statistically significant differences in their distributions between the BZD and PHB groups ($p < 0.001$). *ED Length of Stay* and *ED Length of Stay* also differ significantly (with $p < 0.01$ and $p < 0.05$ accordingly) between the groups. Observe that the *ICU Length of Stay* has median and IQR of 0 in both groups, this is due to majority of patients did not stay in the ICU which will result in 0 length of stay. In particular, there is only one patient in the BZD who got admitted into ICU and has 52 hours of stays and there are 11 patients got admitted into ICU in the PHB group with median and IQR of 69 (40, 266) hours.

Table 6: Outcome Variables: Last and change of CIWA score in ED show a strong significant difference in the BZD and PHB group.

Variable	BZD (N = 1021)	PHB (N = 246)	p value	Test
Primary				
Disposition, no. (%)			< 0.001***	2
Discharged	973 (95%)	31 (13%)	(< 0.001***)	1
Acute (Routine)	36 (3.5%)	156 (63%)	(< 0.001***)	1
Intermediate	11 (1.1%)	47 (19%)	(< 0.001***)	1
Intensive Care	1 (<0.1%)	12 (4.9%)	(< 0.001***)	2
Other Secondary				
Last CIWA score in ED, median (IQR)	5 (2–9)	2 (0–5)	< 0.001***	3
Unknown, no.	274	5		
Change of CIWA score in ED, median (IQR)	-1 (-7 – 0)	-9 (-15 – -3)	< 0.001***	3
Unknown, no.	274	5		
ED Length of Stay (hrs), median (IQR)	15 (8–24)	22 (14–32)	0.011*	3
ICU Length of Stay (hrs), median (IQR)	0 (0–0)	0 (0–0)	0.057	3
Days to ED Presentation, median (IQR)	6 (1–14)	8 (2–15)	0.195	3
Days to Inpatient Readmission, median (IQR)	5 (1–13)	11 (3–19)	0.016**	3
Readmitted within 3 days, no. (%)	22 (5.9%)	4 (4.1%)	0.623	2
Unknown, no.	648	148		
30-day Mortality, no. (%)	1 (<0.1%)	0 (0%)	1.000	2

PHB = phenobarbital group; BZD = non-phenobarbital or benzodiazepine only group; BZD = benzodiazepine; CIWA = Clinical Institute Withdrawal Assessment for Alcohol. Significance codes: '***' $p < 0.001$, '**' $p < 0.01$, '*' $p < 0.05$. Test Label: 1 = Pearson's Chi-Square; 2 = Fisher's Exact ; 3 = Independent Two-sample t. The criteria of the evaluation test is listed in Table 3.

We are also interested in comparing the primary outcome within the PHB group across different dosage subgroups. Due to the small sample size, Fisher's exact test was used to assess differences in distributions. The small p-value (< 0.001) in Table 7 indicates a statistically significant difference.

The p value for each level in disposition come from treating each category as a separate binary outcome (yes/no for being in that category). Each of these tests compares whether the probability of that specific outcome level differs between groups, ignoring the other categories except as the complement. For example, the p-value for Intensive Care tests whether ICU admission versus non-ICU differs between BZD and PHB, rather than testing the full distribution of all disposition types at once. Discharged/ Acute / Intermediate's p-value was obtained using the Chi-squared test on . ICU's p value was obtained using Fisher's exact test due to low count.

The overall p-value for Disposition tests whether the distribution of all disposition categories jointly differs between the BZD and PHB groups. In other words, it asks whether the probability of being in any of the four outcome categories (Discharged, Acute, Intermediate, ICU) is the same across the two groups. This is a single global test of independence between group and the multi-level outcome, typically based on a chi-square or Fisher's exact test for the full contingency table.

Table 7: Primary Outcome in Phenobarbital Group shows highly significant differences for dosage subgroup.

Variable	PHB-Low Does (N = 231)	PHB-High Does (N = 15)	p value
Disposition, no. (%)			0.013
Discharged	26 (11%)	5 (33%)	
Acute (Routine)	151 (65%)	5 (33%)	
Intermediate	44 (19%)	3 (20%)	
Intensive Care	10 (4.3%)	2 (13%)	

PHB = phenobarbital group.

3 Prediction Model

To better understand the factors associated with treatment group assignment and patient outcomes, we used regression models. We used three types of regression models: linear regression, logistic regression, and multinomial logistic regression. These models allow us to evaluate how multiple patient characteristics (such as age, vital signs, and clinical scores) are related to outcomes while accounting for the influence of other variables at the same time.

Linear regression is used when the outcome is a continuous numerical value (such as length of stay or time to readmission). This model estimates how much the outcome is expected to increase or decrease when a predictor variable changes, while holding all other variables constant. In other words, it quantifies the average change in the outcome associated with each variable.

Logistic regression is used when the outcome has two possible categories (such as treatment group: BZD vs. PHB). Instead of modeling the outcome directly, this model estimates the likelihood (or probability) of one outcome relative to the other. It does this by comparing the odds of the outcome occurring across different patient characteristics, while adjusting for other variables in the model.

Multinomial logistic regression extends this approach to outcomes with more than two categories (such as levels of disposition: Discharged, Acute, Intermediate, or Intensive Care). In this model, one category is selected as the reference level (in this study, Discharged), and the model estimates how patient characteristics are associated with the likelihood of being in each of the other categories relative to this reference. This allows us to understand how factors influence not just whether an outcome occurs, but which type of outcome is more likely.

Across all models, the goal is to isolate the independent relationship between each variable and the outcome, while accounting for the presence of other variables. This is particularly important in clinical data, where many factors are interrelated. By using these methods, we can better identify which variables are most strongly associated with treatment selection and patient outcomes.

We also report p-values using a Wald z-test under the null hypothesis of 0 coefficient for the variable, which help assess whether an observed relationship is likely to be real or could have occurred by chance. A p-value in regression model represents the probability of observing a result at least as strong as the one found in the data, assuming that there is actually no true relationship between the variable and the outcome.

Smaller p-values indicate stronger evidence against the idea that there is no relationship. In practice, a p-value less than 0.05 is commonly considered statistically significant, meaning the observed association is unlikely to be due to random variation alone.

However, it is important to note that the p-value does not measure the size or importance of an effect. A very small p-value does not necessarily mean the effect is large or clinically meaningful; it only indicates that the evidence for an association is strong. Therefore, p-values should be interpreted together with effect measures (such as adjusted odds ratios or regression coefficients) to understand both the strength and the practical significance of the results.

3.1 Logistic Regression

For outcomes with two categories (e.g., treatment group: BZD vs. PHB), we used classic logistic regression. For outcomes with more than two categories (e.g., levels of disposition such as Discharged, Acute, Intermediate, or Intensive Care), we used multinomial logistic regression. In multinomial models, one outcome category is selected as the reference level (in this study, Discharged), and all comparisons are made relative to this reference group.

Results from these models are presented as adjusted odds ratios (aOR). The interpretation of aOR depends on the type of variable:

For continuous variables (e.g., age, blood pressure, CIWA score), the aOR represents the change in the likelihood of the outcome associated with a one-unit increase in that variable, while holding all other variables constant. For example, an aOR of 1.05 means that each one-unit increase in the variable is associated with an approximately 5% increase in the likelihood of the outcome.

For categorical variables (e.g., gender, treatment group), the aOR compares each category to a designated reference level. For example, if Female is the reference for gender, an aOR of 2.3 for Male means that males have approximately 2.3 times the likelihood of the outcome compared to females, after adjusting for other variables. Similarly, for race, comparisons are made relative to the reference category (Black/African American in this study).

For the multinomial logistic regression of *Disposition*, each aOR represents the relative likelihood of being in a specific outcome category (e.g., Acute, Intermediate, or Intensive Care) compared to the reference outcome (Discharged). For example, an aOR greater than 1 indicates a higher likelihood of being in that level of care rather than being discharged, while an aOR less than 1 indicates a lower likelihood, holding other variables constant.

In general, an aOR greater than 1 indicates a higher likelihood of the outcome, an aOR less than 1 indicates a lower likelihood, and an aOR close to 1 suggests little or no difference.

3.1.1 Disposition vs. Baseline

Recall that our primary outcome, *Disposition*, consists of four levels ordered by severity: Discharged, Acute (Routine), Intermediate, and Intensive Care (ICU). We first fit a multinomial logistic regression model of using all the baseline variables listed in Table 4 to predict *Disposition*. The aOR and p-values of this regression is listed in Table 8.

Table 8: Adjusted Odds Ratios (Exponentiated Coefficients) which compares the likelihood of each level of care with Discharged and p-values from multinomial logistic regression (Disposition ~ All Baseline Variables).

Variable	Acute (Routine) aOR (p-value)	Intermediate aOR (p-value)	Intensive Care aOR (p-value)
Intercept	0.020 (0.003 **)	0.006 (0.005 **)	≈ 0 (0.982)
Age	1.017 (0.145)	1.018 (0.235)	0.994 (0.887)
Race: Other	1.389 (0.577)	1.530 (0.586)	1.324 (0.848)
Race: White	1.002 (0.997)	1.016 (0.980)	0.530 (0.563)
Gender: Male	0.929 (0.824)	0.898 (0.820)	2.451 (0.501)
Group: PHB	165.50 (<2e-16 ***)	209.14 (<2e-16 ***)	544.03 (1.62e-05 ***)
Subgroup: High Dose	0.047 (0.001 ***)	0.071 (0.012 *)	0.383 (0.551)
Cumulative PHB (mg)	1.002 (0.148)	1.002 (0.212)	0.999 (0.818)
Cumulative BZD (mg)	1.032 (2.74e-08 ***)	1.043 (5.35e-10 ***)	1.054 (3.96e-06 ***)
Route: Intramuscular	0.383 (0.512)	1.785 (0.739)	1.682 (1.000)
Route: Intravenous	2.691 (0.002 **)	6.234 (7.70e-05 ***)	≫ 1 (0.987)
Sedative Med in ED	1.757 (0.064 .)	1.172 (0.694)	1.157 (0.869)
EtOH < 10	1.050 (0.907)	0.311 (0.034 *)	0.220 (0.215)
EtOH (mg/dL)	0.996 (0.062 .)	0.992 (0.007 **)	0.982 (0.094 .)
First CIWA Score	1.001 (0.977)	1.014 (0.603)	1.188 (0.001 ***)
Systolic BP	1.007 (0.445)	1.003 (0.811)	1.024 (0.447)
Diastolic BP	0.976 (0.046 *)	0.979 (0.199)	0.950 (0.178)

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$. ≈ 0 indicates an odds ratio extremely close to 0 due to a very large negative coefficient; ≫ 1 indicates an extremely large odds ratio due to a very large positive coefficient. Reference Level: Discharged (Disposition); Black/African American (Race); Female (Gender); Low Dose (Subgroup); Oral (Route).

By comparing all factors simultaneously in Table 8, we observed the most two significant factors are *Group* and *Cumulative BZD (mg)* with strongly significant p-values ($p < 0.001$) across all three levels, suggesting they are highly correlated with patients' disposition after ED.

Patients in the PHB treatment group were significantly more likely to be admitted to any level of care compared to being discharged. All the aORs for Cumulative BZD are greater than 1.0, it means that as the dose goes up, the likelihood of admission (compared to being discharged) also goes up. There is no 'safe' threshold where the risk jumps; instead, every single milligram added to the patient's treatment in the ED incrementally pushes the needle toward a higher level of hospital admission. To be more specific, for every 1 mg increase in BZD, the odds of being admitted to Acute, Intermediate and Intensive Care is increase by 3.2%, 4.3% and 5.4% accordingly. Although this might look tiny, because this is per milligram, the risk compounds quickly. If you compare a patient who received 10 mg more than another patient, you don't just add the percentages; you multiply the odds (1.032^{10}). For a 10 mg increase: The odds of Acute admission aren't just 3% higher—they are roughly 37% higher ($1.032^{10} \approx 1.37$) than the reference level Discharged.

The *Ordered Route: Intravenous* also shows significant for Acute ($p < 0.01$) and Intermediate ($p < 0.001$) which increase the odd of Acute and Intermediate compare to Discharged for getting the drug through Intravenous than Oral. There are also other factors show significant to some levels of inpatient admission, for example, being in High Dose Subgroup decrease the odd of Acute compare to Discharged by 95.3% with $p < 0.001$ and every 1 increment of First CIWA Score in ED increase the odd of Intensive Care than Discharged by 18.8%.

However, some aOR in Table 8 are unusually small or large. For instance, the large aOR of *Group: PHB* for all three levels indicates that almost everyone in the PHB group was admitted, and almost no one was discharged. This issue is often referred to as a separation when a variable is an almost "perfect" predictor of outcome and overfit to a certain level. This struggle is mainly due to limited amount of data from this retrospective study.

We also perform further variable selection using Lasso and Elastic Net, different combinations of variables suggested by these regularization were chosen to fit the multinomial logistic regression. The coefficient for Lasso and Elastic Net can be found in Table 12 and Table 13 in the Appendix. The aOR and p-values for the model with minimal AIC, which indicates this combination of predictors has a better fit of the model in our dataset, is reported in Table 9. We removed *Systolic BP*, *Gender* and *Sedative Med in ED* in this reduced model and the significant predictors stay the same except for *Diastolic BP*.

Table 9: Adjusted Odds Ratios (Exponentiated Coefficients) and p-values from multinomial logistic regression (Disposition $\sim$ Intersection of Variables from Elastic Net).

Variable	Acute (Routine) aOR (p-value)	Intermediate aOR (p-value)	Intensive Care aOR (p-value)
Intercept	0.032 (0.003 **)	0.006 (0.002 **)	≈ 0 (0.984)
Age	1.019 (0.096)	1.018 (0.227)	1.009 (0.787)
Race: Other	1.41 (0.557)	1.62 (0.537)	1.14 (0.924)
Race: White	1.03 (0.950)	1.05 (0.937)	0.50 (0.517)
Group: PHB	232.0 ($<2e-16$ ***)	304.0 ($<2e-16$ ***)	450.0 ($2.99e-07$ ***)
Subgroup: High Dose	0.087 (0.001 **)	0.126 (0.023 *)	0.368 (0.468)
Cumulative BZD (mg)	1.033 ($8.77e-09$ ***)	1.043 ($1.94e-10$ ***)	1.054 ($5.24e-07$ ***)
Route: Intramuscular	0.41 (0.558)	1.76 (0.755)	1.24 (1.000)
Route: Intravenous	2.79 (0.001 **)	6.50 ($3.61e-05$ ***)	$\gg 1$ (0.987)
EtOH < 10	1.02 (0.968)	0.30 (0.027 *)	0.17 (0.117)
EtOH (mg/dL)	0.997 (0.069)	0.992 (0.006 **)	0.981 (0.053)
First CIWA score	1.00 (0.839)	1.02 (0.485)	1.164 (0.001 ***)
Diastolic BP	0.983 (0.056)	0.983 (0.145)	0.968 (0.183)

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. ≈ 0 indicates an odds ratio extremely close to 0 due to a very large negative coefficient; $\gg 1$ indicates an extremely large odds ratio due to a very large positive coefficient. Reference Level: Discharged (Disposition); Black/African American (Race); Low Does (Subgroup); Oral (Route).

From both models, we see the PHB treatment and higher cumulative BZD exposure are consistently associated with increased likelihood of admission.

3.1.2 Secondary Models

We fit a classical(binary) logistic regression against *Group* using the remaining baseline variables without *Subgroup* (since no patients receive High Does BZD, this is a perfect separation) in Table 10. *Gender*, *First CIWA Score in ED* and *EtOH* shows strong significant ($p < 0.001$) in the model.

Table 10: Adjusted Odds Ratios (Exponentiated Coefficients) and p-values from binary logistic regression (Group $\sim$ Remaining Baseline Variables Without Subgroup).

Variable	Adjusted Odds Ratio (aOR)	p-value
Intercept	0.472	0.328
Age	0.997	0.626
Race: Other	1.545	0.196
Race: White	1.474	0.133
Gender: Male	2.311	$3.67e-05$ ***
Weight (kg)	0.989	0.014 *
Systolic BP	0.989	0.045 *
Diastolic BP	1.012	0.068 .
First CIWA Score in ED	1.056	$4.71e-07$ ***
EtOH (mg/dL)	0.997	0.001 ***
EtOH < 10	0.893	0.622

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$. Reference Level: Black/African American (Race); Female(Gender).

Recall that the distributional difference analyses identified several variables that differed significantly between the BZD and PHB groups, including race, gender, CIWA score, ethanol measures, and diastolic blood pressure. However, in the logistic regression model, only a subset of these variables—most notably gender, CIWA score, and ethanol level—remained independently associated with treatment group, while others such as race were no longer significant after adjustment. Additionally, variables such as weight and systolic blood pressure, which were not significant in univariate comparisons, emerged as significant predictors in the adjusted model.

These differences reflect the distinction between univariate distributional analyses and multivariable modeling. Distributional tests capture overall group differences but do not account for relationships between covariates, whereas logistic regression isolates the independent effect of each predictor while controlling for potential confounding. As a result,

some variables lose significance after adjustment due to shared variance with stronger predictors, while others become significant once confounding is accounted for.

We also fit a classical(binary) logistic regression against *Subgroup* using the remaining baseline variables without *Group* but no predictors shown statistical significant. The detail of this model can be found in Table 7 in Appendix.

Other categorical second outcomes such as *Readmitted Rate within 3 days* and *Mortality Rate within 30 days* are also fit against baseline variables in binary logistic regression. No significant are found from the model for *Readmitted Rate within 3 days*, see Table 15 in Appendix. The algorithm to fit against *Mortality Rate within 30 days* does not converge due to lack of data and separation.

3.2 Linear Regression

We fit the remaining secondary outcomes using linear regression and elaborate the model for the *Change of CIWA in ED* first.

We examined how patient characteristics and treatments are associated with changes in CIWA score during the emergency department stay (measured as the difference between the last and first CIWA scores). A negative change indicates improvement in symptoms, while a positive change indicates worsening. The model suggests that several factors are significantly associated with changes in CIWA score, although much of the variation remains unexplained (R-squared: 0.2075; approximately 20% of the variation is captured by the model). The coefficient and p-values can be found in Table 11.

Table 11: Linear Regression Results for Change in CIWA Score (Last - First).

Variable	Coefficient (β)	Std. Error	p-value
(Intercept)	2.640	2.205	0.232
Age	-0.007	0.019	0.712
Race: Other	-0.595	1.006	0.555
Race: White	0.195	0.707	0.783
Gender: Male	1.205	0.570	0.035 *
Group: PHB	-4.830	0.806	<0.001 ***
Subgroup: High Dose	1.990	2.093	0.342
Route: Intramuscular	1.547	1.603	0.335
Route: Intravenous	-2.250	0.539	<0.001 ***
Received Sedative in ED	0.141	0.515	0.784
Cumulative PHB (mg)	-0.002	0.002	0.268
Cumulative BZD (mg)	-0.043	0.010	<0.001 ***
EtOH < 10 mg/dL	1.557	0.736	0.035 *
EtOH (mg/dL)	0.013	0.003	<0.001 ***
Systolic BP	-0.026	0.017	0.117
Diastolic BP	-0.033	0.021	0.111

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

$R^2 = 0.2075$; Adjusted $R^2 = 0.1936$.

Note: A negative coefficient indicates a greater reduction in CIWA score from admission to disposition.

After adjusting for all other variables, patients in the PHB group showed, on average, a 4.83-point greater reduction in CIWA score compared to the BZD group, indicating substantially greater symptom improvement. Additionally, patients who received medication via the intravenous route had, on average, a 2.25-point greater reduction in CIWA score compared to those who received oral treatment.

In contrast, higher cumulative BZD dose was associated with worsening outcomes: for each additional 1 mg of BZD administered, the CIWA score increased by approximately 0.043 points on average. Similarly, ethanol level was positively associated with CIWA change, with each 1 mg/dL increase in ethanol corresponding to an increase of about 0.013 points in CIWA score. Patients with very low ethanol levels (EtOH < 10) had, on average, a 1.56-point higher CIWA change compared to others.

Male patients had, on average, a 1.20-point higher CIWA change compared to female patients, indicating slightly less improvement. Other variables, including age, race, blood pressure, cumulative PHB dose, and use of additional sedative medication, were not statistically significant predictors in this model.

In summary, treatment group, route of administration, cumulative BZD dose, and ethanol levels appear to be the most important factors associated with symptom changes, while many other patient characteristics showed limited influence in this model.

We also perform similar study for *ED Length of Stay* and *Hospital Length of Stay*, refer Table 16 and Table 17 in the Appendix for details.

4 Conclusion

This study provides an evaluation of treatment patterns and outcomes for alcohol withdrawal in the emergency department using both descriptive and regression approaches. Across multiple models, treatment group, cumulative benzodiazepine dose, and route of administration consistently emerged as the strongest predictors of patient disposition and symptom progression. Patients receiving phenobarbital demonstrated greater improvement in CIWA scores but were also substantially more likely to be admitted, likely reflecting confounding by indication, where more severe cases preferentially receive phenobarbital. Higher cumulative benzodiazepine exposure was consistently associated with increased likelihood of admission and poorer symptom trajectories, suggesting that escalating dosing may reflect or contribute to clinical severity.

However, all findings in this study represent statistical associations rather than causal relationships; due to the retrospective and observational nature of the data, the models identify relationships after adjustment for covariates but cannot establish causation, and residual confounding may still be present. To investigate further in the causal relationship, a randomized, double-blind controlled trial would be necessary.

Similar results were suggested by the regularization method which provide more evidence of our result. While penalized methods improved model stability and supported variable selection, issues such as separation and limited subgroup sample sizes highlight the constraints of retrospective observational data. Overall, these findings underscore the importance of accounting for treatment selection bias and suggest that both medication choice and dosing intensity play critical roles in clinical outcomes in alcohol withdrawal management.

AI Use

AI tools (ChatGPT) were used to assist in refining the clarity, structure, and readability of this report. AI was used to assist in LaTeX to create Tables. I also consult AI for unfamiliar packages in R.

Appendix

A Additional Tables

Table 12: LASSO Regression Coefficients at Optimal Lambda ($\lambda_{min} = 0.0067$) for Patient Disposition.

Variable	Discharged	Acute (Routine)	Intermediate	Intensive Care
(Intercept)	4.4157	0.6399	-0.8141	-4.2414
Age	-0.0106	0.0006	.	.
Race: Other	-0.1018	.	.	.
Race: White	.	.	.	.
Gender: Male	.	.	.	.
Group: PHB	-4.7700	.	.	.
Subgroup: High Dose	1.2951	-0.6956	.	.
Route: Intramuscular	.	-0.1022	.	.
Route: Intravenous	-0.6729	.	0.3821	.
Cumulative PHB (mg)	-0.0008	.	.	.
Cumulative BZD (mg)	-0.0281	-0.0036	0.0036	0.0050
Sedative Med in ED	.	0.3332	.	.
EtOH < 10	.	0.2072	-0.1192	.
EtOH (mg/dL)	0.0021	.	.	.
First CIWA Score	.	-0.0007	.	0.1193
Systolic BP	.	.	.	.
Diastolic BP	0.0078	-0.0003	.	.

Note: Coefficients are reported at the λ_{min} value determined by cross-validation. A dot (.) indicates that the coefficient was shrunk to zero by the LASSO penalty, effectively removing the variable from the model for that specific category.

Table 13: Elastic Net Regression Coefficients at Optimal Lambda ($\lambda_{min} = 0.0102$) for Patient Disposition.

Variable	Discharged	Acute (Routine)	Intermediate	Intensive Care
(Intercept)	4.1413	0.6008	-0.8315	-3.9105
Age	-0.0095	0.0019	.	.
Race: Other	-0.1655	.	.	.
Race: White	.	0.0043	.	.
Gender: Male	.	.	.	.
Group: PHB	-4.0620	0.3400	0.2203	.
Subgroup: High Dose	1.2170	-0.9166	-0.2282	.
Route: Intramuscular	.	-0.3390	.	.
Route: Intravenous	-0.6215	.	0.4069	0.2093
Cumulative PHB (mg)	-0.0016	.	.	.
Cumulative BZD (mg)	-0.0227	.	0.0072	0.0092
Sedative Med in ED	.	0.3401	.	.
EtOH < 10	.	0.2120	-0.1824	.
EtOH (mg/dL)	0.0022	.	.	.
First CIWA Score	-0.0032	-0.0062	.	0.1002
Systolic BP	.	.	.	.
Diastolic BP	0.0078	-0.0012	.	.

Note: Coefficients are reported at the λ_{min} value determined by cross-validation. The Elastic Net penalty allows for both variable selection (setting coefficients to zero, represented by a dot) and the grouping of highly correlated predictors.

Table 14: Adjusted Odds Ratios (Exponentiated Coefficients) and p-values from binary logistic regression (Subgroup ~ Remaining Baseline Variables Without Group).

Variable	Adjusted Odds Ratio (aOR)	p-value
Intercept	≈ 0	0.990
Age	1.002	0.936
Race: Other	$\gg 1$	0.995
Race: White	$\gg 1$	0.996
Gender: Male	0.897	0.880
Weight (kg)	0.995	0.752
Systolic BP	1.010	0.628
Diastolic BP	1.036	0.192
Group: PHB	$\gg 1$	0.991
First CIWA Score in ED	1.048	0.122
EtOH (mg/dL)	0.999	0.792
EtOH < 10	0.887	0.883

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$. Reference Level: Black/African American (Race); Female(Gender); BZD(Group).

Table 15: Adjusted Odds Ratios (aOR) and p-values for Readmission within 3 Days.

Variable	Adjusted Odds Ratio (aOR)	p-value
(Intercept)	2.179	0.751
Age	1.001	0.949
Race: Other	0.626	0.605
Race: White	0.390	0.131
Gender: Male	2.004	0.304
Group: PHB	2.436	0.565
Subgroup: High Dose	≈ 0	0.997
Route: Intramuscular	≈ 0	0.995
Route: Intravenous	1.300	0.646
Received Sedative in ED	1.298	0.640
Cumulative PHB (mg)	0.987	0.276
Cumulative BZD (mg)	1.012	0.130
EtOH < 10	1.079	0.924
EtOH (mg/dL)	0.997	0.440
First CIWA Score	0.942	0.164
Systolic BP	0.975	0.190
Diastolic BP	1.002	0.926

Reference Level: Discharged (Disposition); Black/African American (Race); Low Does (Subgroup); Oral (Route).

Table 16: Linear Regression Results for Emergency Department Length of Stay (Hours).

Variable	Coefficient (β)	Std. Error	p-value
(Intercept)	29.364	5.240	<0.001 ***
Age	0.117	0.046	0.011 *
Race: Other	0.888	2.381	0.709
Race: White	-2.008	1.675	0.231
Gender: Male	-0.379	1.350	0.779
Group: PHB	-5.917	1.914	0.002 **
Subgroup: High Dose	-11.343	4.957	0.022 *
Route: Intramuscular	9.287	3.800	0.015 *
Route: Intravenous	-5.953	1.305	<0.001 ***
Received Sedative in ED	7.673	1.219	<0.001 ***
Cumulative PHB (mg)	0.040	0.005	<0.001 ***
Cumulative BZD (mg)	0.303	0.024	<0.001 ***
EtOH < 10 mg/dL	-0.657	1.743	0.706
EtOH (mg/dL)	-0.010	0.007	0.173
First CIWA Score	-0.598	0.086	<0.001 ***
Systolic BP	-0.024	0.039	0.541
Diastolic BP	-0.088	0.049	0.072 .

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$.

$R^2 = 0.3058$; Adjusted $R^2 = 0.2929$.

Note: Coefficients represent the change in hours of ED stay for every unit increase in the predictor.

Table 17: Linear Regression Results: Predictors of Total Hospital Length of Stay (Hours).

Variable	Coefficient (β)	Std. Error	p-value
(Intercept)	-20.385	33.182	0.539
Age	0.252	0.291	0.388
Race: Other	35.763	15.081	0.018 *
Race: White	-2.367	10.607	0.823
Gender: Male	4.520	8.547	0.597
Group: PHB	118.793	12.124	<0.001 ***
Subgroup: High Dose	-27.467	31.393	0.382
Route: Intramuscular	4.995	24.066	0.836
Route: Intravenous	12.732	8.267	0.124
Received Sedative in ED	8.375	7.722	0.278
Cumulative PHB (mg)	-0.015	0.032	0.649
Cumulative BZD (mg)	0.938	0.155	<0.001 ***
EtOH < 10 mg/dL	8.105	11.036	0.463
EtOH (mg/dL)	-0.011	0.045	0.811
First CIWA Score	0.103	0.543	0.849
Systolic BP	-0.053	0.249	0.830
Diastolic BP	-0.219	0.309	0.479

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

$R^2 = 0.2388$; Adjusted $R^2 = 0.2246$.

Note: Coefficients represent the change in total hospital hours for every unit increase in the predictor.